Bhanu Agarwal

M.Sc. NLP Student | Python, SQL, Semantic Search

Trier, Germany
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Education

04/2025 – present **M.Sc. Natural Language Processing**, University of Trier, Germany
Current average: 1.7 across 50 ECTS of passed examinations.

10/2023 – 03/2025 **M.Sc. Data Science**, University of Erlangen-Nürnberg (FAU), Germany
Transferred to Trier to specialise in Natural Language Processing.

07/2019 – 05/2023 **B.Tech. Computer Science and Engineering**, Vellore Institute of Technology, India
Final grade: 8.22/10

Work Experience

01/2023 – 03/2023 **Software Development Intern**, Capella Infotech Solutions, Chennai, India

- Fixed software bugs and implemented logic specified by senior developers.
- Labelled and segmented customer reviews for a sentiment-classification project.

Selected Projects

JobFit Navigator | GitHub: jobfit-navigator | Live demo
Python, Streamlit, SQLite, BM25, Sentence Transformers

- Built a Streamlit and SQLite workspace for job listings, profile evidence and source-linked review of supported English and German requirement wording.
- Combined BM25 and pretrained multilingual embedding search over overlapping passages using reciprocal rank fusion (k=60), with metadata filtering before ranking.

Local RAG for Job Descriptions | GitHub: job-description-rag
Python, Ollama, ChromaDB, Streamlit

- Built a local RAG application over 40 job descriptions using nomic-embed-text embeddings, ChromaDB and Llama 3.2 3B, with answers linked to inspectable source passages.
- Reviewed retrieval, answer correctness and citation support on 15 development questions with saved contexts and 6 fresh end-to-end questions; validated response structure and source IDs.

Technical Skill Extraction from Job Descriptions | GitHub: alias-aware-technical-skill-extraction
Python, regular expressions, evaluation

- Built strict and alias-aware lexicon matchers for technical skills in English IT job descriptions.
- Used a reproducible 200/100 split, a frozen 20-label lexicon and manually annotated gold labels; raised micro-F1 from 0.862 to 0.984, recovering 80 of 92 baseline false negatives.

Technical Skills

Programming and tools: Python; SQL; Git/GitHub; Streamlit, SQLite, Docker

Retrieval and NLP: BM25, pretrained multilingual embeddings, semantic and hybrid search; text preprocessing, lexicon design; Hugging Face Transformers; RAG

Data and ML: pandas, NumPy, scikit-learn, PyTorch; precision/recall/F1 and ROC-AUC; Matplotlib, Seaborn, Excel

Languages & Availability

Languages English: C1 | German: A1

Availability Available for internship and working-student opportunities across Germany.